Epileptic Seizure Prediction using Multi-Channel Vision Transformer

Chandan Kumar and Proff. R.K Aggarwal†

[1*]M.tech(Computer Engineering), Nit Kurukshetra, Kurukshetra, 136119, Haryana, India.
[2]M.tech(Computer Engineering), Nit Kurukshetra, Kurukshetra, 136119, Haryana, India.

*Corresponding author(s). E-mail(s): chandankumar08032001@gmail.com;
Contributing authors: rka15969@gmail.com;
†These authors contributed equally to this work.

Abstract

-Epilepsy is a disease of the nervous system, manifested by irregular, sudden seizures, and often loss of consciousness. Even though antiepileptic medications are accessible, only 30% of epilepsy suffer chronic seizures. Thus, it stresses the relevance of precise seizure prediction for patients and their carers to avoid triggers and prevent the risk of injuries during these attacks.This paper presents a new Transformer-based model, Multi-channel Vision Transformer (MViT), which aims at directly identifying multi-channel temporal-spatial-frequency features from EEG data. The Continuous Wavelet Transform (CWT), an easy but highly efficient method for transforming time series EEG data into a scalogram(a graphical representation in the time frequency domain)is used in the preparation step. These scalograms are divided into patches of a certain size that do not overlap and serve as input to MViT for the EEG signal's classification. The studies conducted on two common EEG datasets provide strong support for the suggested MViT strategy. According to the data, MViT performs better than other seizure prediction techniques and has a sensitivity of 90.28–91.15% for invasive EEG and 99.80% for surface EEG.

Keywords: EEG; epilepsy; seizure prediction; continuous wavelet transform; vision transformer

1 Introduction

Epilepsy is a neurological disorder that affects the brain and which is classified by the irregular discharge of electricity in the brain. It expresses itself through seizures that are random and may comprise of factors like spasms, changes of consciousness for instance, vacuuming, and in some instances passing out.As of now, AEDs(anti-epileptic drugs) act as the main management approach in epilepsy treatment.However, approximately 30% of epilepsy patients continue to experience seizures despite treatment, while the remaining 70% achieve seizure control through medical intervention.The patients continue to suffer from the side effects of these medications such as stomach pain, fatigue, dizziness, and blurred vision all of which compromise the patient's quality of life. If patients fail to respond to medication then they may be considered for epilepsy surgery.Jul 3 2012 Mishandling prescriptions may lead to epilepsy surgery being considered. This operation entails the removal or disconnection from the brain area for controlling or eliminating seizures. But severe complications are expected after this surgery: vision loss, disorders of memory and language, stroke, paralysis. Such difficulties have contributed to the rising need for techniques and strategies in the prediction of seizures among the researchers[1][2]. Epileptic seizure prediction presents a great opportunity of letting the patient know that a seizure is about to occur so they can avoid situations that may cause further injury and give them time to take a quick acting drug.At present, the electroencephalography(EEG) is the most commonly employed method for surveillance of seizures and their forecasting. The first problem lies in the fact that it is necessary to differentiate pre-seizure EEG activities by recognizing certain activity patterns that denote an oncoming seizure, which is the general aim of the proposed study.
Epilepsy researchers have classified brain EEG activity in individuals with epilepsy into 4 primary states: proseizure (before the seizure occurs), periclinal (during the seizure), postseizure (after the seizure occurs) and the time between consecutive seizures known as, intrasiezure. Therefore, to use this approach successfully for seizure prediction a desirable algorithm must be able to capture brain activity during preictal phase to predict seizures and distinguish it from interictal phase, which is normal brain activity and is a false positive. In order to overcome this challenge, several approaches of feature extraction and classification techniques for preictal and interictal EEG signals are proposed.

The paper presents a novel transformer-based technique for the classification of preictal and interictal EEG activity of high accuracy. The main findings of this study involves the application of Continuous Wavelet Transform (CWT), which transforms the time-series EEG data into image-like descriptors called scalograms. These scalograms retain local frequency and time characteristics of the EEG and integrity of features is crucial for the analysis of the signal. MViT is proposed to take the full advantages of multi-channel nature of EEG data, and performs multi-channel temporal-spectral feature extraction for superior classification performance with the MViT model that has the ability to model temporal-spectral patterns both for each channel and the concatenated feature series of all the channels. Furthermore, the comparative experiments were also performed to test the effectiveness of the proposed methods.

2 Related Work

Several seizure prediction methods have been established in the last 10 years using machine learning and signal processing algorithms. Before deep learning became popular, these techniques usually followed a traditional pipeline with two steps: first, feature engineering was used to find distinct EEG features about the various classes of brain activity, and then, either traditional or modern machine learning algorithms were used. This method was applied to a seizure advisory system (SAS), which uses data from invasive EEG (iEEG) to anticipate seizures in advance[3]. The prediction sensitivity of the method, which was tested on 15 adult patients with drug-resistant epilepsy, ranged from 17% to 100%; interindividual variability was significant. In another seizure prediction method, Kiral-Kornek et al.[4] employed the method using the intracranial EEG data of ten patients in the SAS trial. A quick survey of the latest seizure prediction finds that frequency and time-frequency features are some of the used behaviors when extracting the attributes to analyze the EEG data. Moreover, the traditional classifiers have been more or less sidelined by the SVM as the most popular classifier algorithm in this field. For instance, a study presented an algorithm for an attempt of predicting seizures from common features present on the EEG frequency rhythms and powerful. Hence, a new cost-specific SVM for treating the structurally imbalanced interictal and preictal samples was designed and trained with 12 patients from the Freiburg EEG database, and validated on the additional 6 patients, resulting in an average sensitivity of 97.5% of the developed model as well as 0.27 false alarms per hour. The same analysis of the EEG features using frequency domain and support vector classifier with other data sets[5][6][7] resulted in prediction sensitivities between 90% to 92%. Williamson et al.[8] stressed on the use of concurrent EEG m-features rather than the simple Poincaré like features with which multivariate EEG analysis is often restricted. They fed the SVM classifier with multivariate EEG features; their cross-validation on the Freiburg EEG database for 19 patients provided them with an AUC that ranged from 0.936 to 0.972.

Recently, deep learning has appeared and brought the possibility of further steps in the epileptic seizure prediction field. Fear of failure and history of seizures do not seem to be risk factors for the development of seizures at the current time, thus directly predicting seizures from neural signals has been made easy with the end-to-end learning utilising deep neural network where feature extraction and selection has been done automatically by the respective software.. With regard to time-frequency representations the following deep learning approaches have been designed: Spectrogram-based architectures and others.In the current study, the CNN mentioned in [9] with a sequence of six convolutional layers and two fully connected layers was applied in differentiating between interictal and preictal EEG activities in two independent scalp EEG databases. This method had a sensitivity of 87.8% and the false prediction rate recorded per hour was 0.142P using 3D wavelet tensors. Another method described in [10] involved changes of the raw EEG signals to time-frequency images through application of the STFT. Such images were then passed through a CNN model with three layers of convolution and two layers of fully connected layers. Alternative evaluation methods for the model were employed through aggregating data from Freiburg

Hospital CBH-MIT and AES seizure forecast competition datasets, where the values of sensitivity harness average of 81.4%, 81.2% and 75% for executing distinction among patients at risk of seizures. Similar to Wang et al.[11] CNN model developed has three of each of the convolutional and the fully connected layer. The training of this model was done on feature maps generated by channel-frequency using directed transfer function. By applying the method on the Freiburg EEG dataset it achieved a sensitivity of 90.8% and had a false prediction per hour of 0.08.

Epileptic Seizure prediction is one of the least studied areas, although there is a somewhat diverse range of CNN architectures implemented to this end. The 2D-CNN models have been used to evaluate the CHB-MIT dataset[12] In our case, Ozcan et al. [13] used a method based on analysis of 3D-CNN technique they involved time and frequency analysis of the EEG data, with sensitivity of 85.7%. Liu et al.[14] have also designed a multi-view CNN architecture which can process both the frequency and time views of a given input. Specific techniques used to comprise time domain features included applying FFT directly on raw EEG to obtain frequency domain features, autoregression coefficients, correlation, and signal entropy. These features were passed through convolutional layers independently and an embodiment of the features was done through a fully-connected layer. The model acquired an AUC of of 0.837 on the AES invasive EEG database and AUC of 0.82 and 0.89 for two subjects in CHB-MIT surface EEG database. To solve the domain-shift issue between the training and the testing images, a new architecture of CNN with multi-scale temporal convolution was introduced in [15]. This model eventually captured preictal features over different time intervals thus improving the reliability and comparability of features across the phases of seizure forecasting. Relative to the previous approach, this method exhibited improved insensitivity to fluctuations in signal characteristics, translating to about 9% increase in seizure prediction accuracy. These improvements were assessed on the utilised public datasets, the EEG database and the AES Kaggle the effectiveness of the model proves its reliability on multiple datasets.

Hussein et al.[16] proposed a semi-dilated convolutional network for action recognition that essentially increases the sizes of receptive fields of convolution filters in temporal domain while retaining high-frequency information. Similarities were applied and realized on EEG scalograms already with 98.8% accuracy of the seizure prediction on the public scalp EEG database and within the range of 88.5-89.52% for two invasive EEG databases proved its effectiveness in different applications. Subsequently, graph convolutional networks (GCNs) have been used in seizure prediction. The GCN-based methods map the EEG channels to the nodes of the graph and the interrelations of the nodes as the edges to construct the multi-channel EEG graph topological. Lian et al.[17] presented a novel graph structure to jointly model global and local relationships between the EEG signals and introduced an additional representation learning network that improves the understanding of the inter-channel interactions.In experiments conducted on the Freiburg EEG dataset, this network has shown a better performance in compared with other classifiers including SVM, XGBoost and 2D-CNN in predictive accuracy and sensitivity and, therefore, can be considered as a reliable seizure prediction method.

3 Datasets and Methodology

3.1 Datasets

The performance of the proposed Multi-channel Vision Transformer (MViT) model for representation learning on the scalp and invasive EEG data of human and animal subjects including pediatric and adults as well as canines is assessed. The above evaluation based on public datasets also allows for MViT model comparison with the current state-of-the-art seizure prediction techniques.

CHB–MIT Scalp EEG Dataset [18]—Recently, the CHB-MIT dataset was obtained from EEG data of pediatric patients with drug-resistant epilepsy in Children's Hospital Boston. These recordings were made after several days of video-monitoring after withdrawal of anti-epileptic drugs prior to epilepsy surgery. The dataset includes EEG data from 22 patients: There were five males, aged from 3 to 22 years and 17 females aged from 1.5 to 19 years; speech samples were recorded at a vocal sampling rate of 256 Hz and 16 bits. On average each patient provided 9 to 42 consecutive one-hour long scalp EEG recordings. Most participants had their electrode placement done by the international 10–20 system which utilized 23 head electrodes to obtain multi-channel EEG data. Less frequently, 24 or 26 electrodes were used to obtain denser data. In the CHB-MIT dataset, 198 seizures are recorded, and the corresponding onset and termination time annotations are available with EEG measurements.

Invasive EEG Dataset of Kaggle/Melbourne University[19]-The intracranial EEG data for this study was directly obtained from three adults with drug-resistant focal epilepsy, using the NeuroVista Seizure Advisory System as described by[15]. The first patient was a 22 year old female who had a primary diagnosis of focal epilepsy since the age of 16. She was on several anti-epileptic drugs over the period in which she was taking part in the trial and she had epilepsy surgery before taking the trial. The second patient was a 51-year-old woman who was diagnosed with focal epilepsy the age of 10 and she was taking Carbamazepine during the study. The third patient, a 50-year-old woman, had frontal lobe epilepsy diagnosed at the first time in the age of 15 and had the surgery before the trial. In total they made 16 electrode placements on the surface of their brains with a concentration on what the researchers believed to be the generators of seizure activity. These electrodes were linked with a wireless module which was surgically installed beneath the skin at the subclavicular region. The signals were sampled at 400Hz with an analog to digital conversion of 16 bits and data acquired was transferred to a handheld device and stored in a flash.

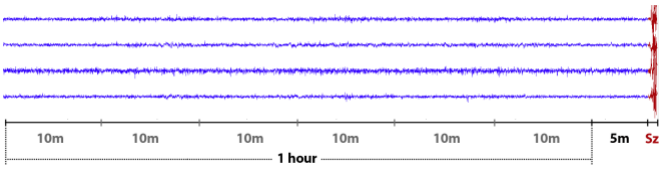

Fig. 1 Examples of one-hour preictal (pre-seizure) EEG signals with a 5-min offset before seizures; Sz denotes the seizure onset. For convenience, only four channels are plotted.

3.2 Methodology

The multi-channel vision transformer that is discussed in this paper extends from the work done by Dosovitskiy et al.[20] where they develop a strong foundation on recognition tasks of images. Building from this, our model proposed a multi-path design optimized for extracting multi-channel EEG characteristics to improve the seizure prediction. This architecture comprises several limbs that separately analyze one EEG channel at a time. These branches run in parallel while learning and updating their tempo-spectral features important for correct and reproducible EEG classification.

3.2.1 EEG Pre-Processing

The proposed EEG pre-processing involves two distinctive steps: (1) **EEG segmentation**, which partitions the time-series EEG recording into smaller non-overlapping segments, and (2) **encoding these segments** into image-like format by adopting Continuous Wavelet Transform (CWT). It must maintain temporal and spectral aspects of the data and therefore is suitable for feature extraction.

EEG Segmentation: Furthermore, it is noteworthy that both surface and intracranial EEG examined in this study is non-stationary, which means that its characteristics evolve over time . The first goal of EEG segmentation is to divide such signals into shorter pseudo-stationary segments for which statistical properties are more stable. Furthermore, segmentation increases the number of labeled data samples much more than accounting for better vision transformer efficiency. In this study, these baselines are measured in terms of 10 minutes clips of EEG where each clips is segmented into 60 segments with 10 seconds each and in without overlapping. This segmentation method leads to a sixty-fold increase in the total number of both interictal and preictal EEG samples, the dataset for model construction effectively becomes much larger.

Mapping EEG Segments into Images: Many of the parameters used for the automatic classification of different brain state of the EEGsignalatics are clinically useful for seizure detection and prediction. Whereas in the previous works the temporal or spectral EEG features are manually extracted, this sometimes produces a large number of false positives. To deal with this, wavelet transform which is a stable time-frequency analysis tool offer a combination of temporal and spectral representation called scalograms.Following the segmentation of the EEG signals, the Continuous Wavelet Transform (CWT) is implemented on each 10s EEG segment to produce two dimensional scalogram images.The scalograms are then taken to the MViT model as network Inputs which are described below. For example, a 10 s preictal EEG recording at 400 Hz is converted to the power spectrum density function. The data is further visualized in 3D and 2D formats with more emphasis based on the temporal and frequency aspects. The resulting scalograms show different patterns of interictal and preictal states, with interictal waves significantly weaker in terms of power scale within similar time-frequency ranges. The characteristic differences noted are effectively utilized in improving the scalogram images for a better and precise seizure prediction system.

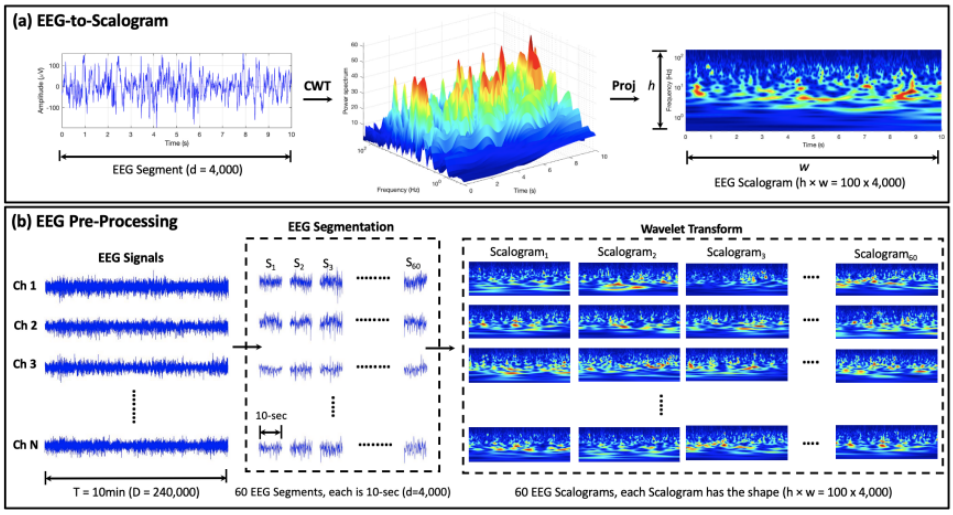

Fig. 2 The proposed pre-processing strategy of EEG for seizure prediction has following two important steps. First, scalograms are obtained using the CWT method that changes the EEG data into power spectrum form. After that, the resulting 3D power spectrum is transformed into a 2D scalogram.

The EEG data maintains a structure of $M \times N \times D$, in which the number of samples of the EEG data is represented as M, the total number of channels for scalp and invasive EEG signals are respectively 23 and 16 while the number of time frames in the 10-minute EEG clip is D which is 153600 for scalp EEG and 240000 for invasive EEG. Segments reached the final shape of $60M \times N \times d$, where 60 refers to the number of 10-second segments in every 10-minute video, and d is the number of data points in each 10-second segment (2560 for scalp EEG, 4000 for invasive) When applying CWT in each EEG channel, the data shape changes its form to $60M \times N \times h \times w$, where h and w is the height and width of the scalogram images of the EEG. Discussed in the following section, these transformations, shown in Figure 2b, help ensure that the raw EEG data is ready for representation learning and classification using MViT.

3.2.2 MViT for EEG Representation Learning

The Transformer model which was designed for natural language processing tasks provides good performance improving which uses MLP layers on top of the multi-head attention for the long-term sequential relationships learning [21]. Recently, the Vision Transformer (ViT) has been demonstrated on different computer vision tasks such as image classification and segmentation[. Based on these progresses, we introduce the Multi-Channel Vision Transformer (MViT), which takes EEG data from multiple channels. More precisely, the MViT has branches, and every branch deals with a different EEG scalogram image in particular. This information is then fused and employed in EEG classification in an attempt to better capture representation from multi-channel EEG data.

The suggested MViT approach is depicted in figure 3 below. The model includes N transformer encoders where each one is fed with one of the N scalogram images that are obtained from the EEG sequences.For each 2D scalogram image of size H $\times$ W, fixed spatial patches of non-overlapping segments $x_p \in \mathbb{R}^{L \times P^2}$ are chosen given that P $\times$ P is the patch size. The patches are flattened, which is followed by projecting them into lower-dimensional vectors known as patch embeddings through linear projection. We have the patch embeddings thus, with the dimension being D, that is, the same with the latent vector used in all the transformer layers.

Due to the fact that positional information is crucial in computer vision tasks, position embeddings are integrated with the patch embeddings. This produces the sequence of patches which is then augmented with positional information to create the final input to the transformer encoder block. The initial transformer encoder used in this work is based on the general transformer architecture described in [21], and it is used to decode veterinary scalograms and acquire their representations. The architecture of the transformer encoder allows for using multi-head self-attention (MSA) to capture both the local and global patterns of the input images and a multi-layer perceptron (MLP) for folding features of the same size. Before every block, there is an implementation of layer.

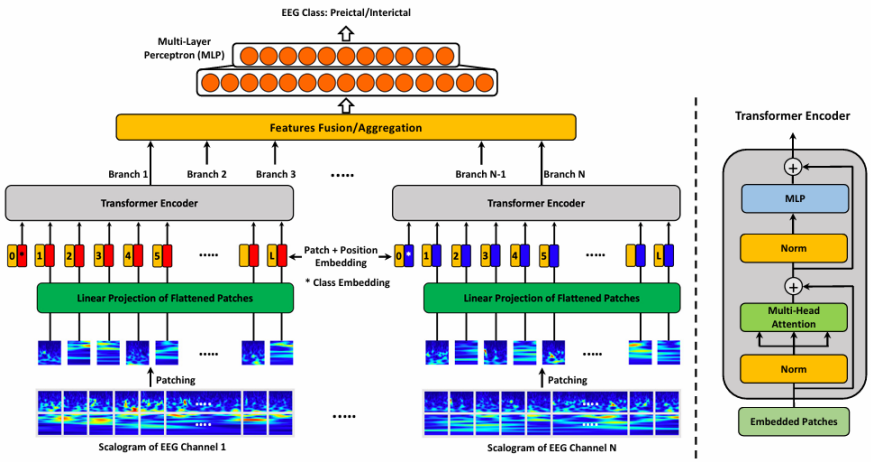

Fig. 3 Structure of MViT for multi-channel EEG feature learning.The architecture consists of N transformer encoders, each processing image tokens from a specific EEG channel. The extracted output feature representations are fused and passed as input to the MLP for classifying EEG signals.

3.2.3 Performance Evaluation

The effectiveness of the applied MViT model is tested to predict real-time seizures via accuracy (ACC%), sensitivity (SENS%), specificity (SPEC%), false-positive rate (FPR), and the area under curve ROC (AUC). They sum up the criteria that would make the model useful in detecting seizures and offer valuable insight into the quality of signals while excluding interfering signals and differentiating between various states of EEG.

4 Results and Discussion

The effectiveness of MViT for seizure prediction is assessed in the proposed method and compared to that of other recent and earlier methods using surface and invasive EEG datasets.

4.1 MViT Prediction Performance on Invasive Human EEG

In this section, we apply the developed seizure prediction algorithm on invasive EEG data from the Kaggle/Melbourne University Seizure Prediction dataset. Seveny individual spike and sharp wave recordings are presented from a total of 3 adult females who are either drug resistant epilepsy or had epilepsy surgery before being included in this clinical trial. For the comparison, we have included our MViT model's results with several other approaches which got the best position in Kaggle competition[22] alongside several other fundamental machine learning algorithms [[3], [4], [16], [23]]. In [3], Cook et al. presented the first in-human seizure advisory system to assist patients with drug-resistant epilepsy. Post electrode implantation their method further grouped time intervals into low, medium and high probability of seizure. Taken together with the latter, the overall average prediction sensibility by the model across the patients was 61.20% while the three patients in this set showed a much lower sensibility at an average of 33.67%.This underperformance was due to drift in temporal features of the EEG data, meaning that the occurrence of patient status predictions was less reliable for these patient types.The prediction problems for such patients are key to the realization of seizure prediction as a viable solution for patients with temporal variability in their EEG signals. These limitations can be addressed using our newly proposed multi-channel based MViT model that has better and rather enhanced features representation inclusions.

In [24], Karoly et al proffered a method of circadian seizure forecasting for the purpose of detecting preseizure brain activities. They suggested applying the value of spike rate identified in preictal EEG records as an objective marker of an oncoming seizure. However, this biomarker had poor performance and could not be tested on other patients. Specifically, the spike rate increased prior to seizures for 9 patients but decreased for the other 6.For evaluating the temporal feature engineering, logistic regression was used, best overall prediction accuracy values were achieved: 62.10% for all fifteen patients including the present study and 52.67% for the sensitivity of three patients in the present study. More recently, Kiral-Kornek et al. [4] proposed

the idea of developing personalized electroencephalogram seizure detectors based on deep learning. Their work showed that de-personalized seizure prediction is possible and that personalization makes the accuracy of the prediction significantly better.This approach enhanced the prediction of seizures in most of the patients bringing the overall average prediction sensitivity to 69.00% for the fifteen patients. Thus, using their method for the three patients analysed in the current study, the sensitivity derived was 77.36%. On the other hand, our proposed method incorporates a multi-channel vision transformer that aims at detecting previously undetectable pre-seizure patterns in the EEG signals. Compared with prior approaches, our model overcomes the shortcomings with the help of the higher quality feature extraction and feature representation learning.

In the same context, we also compare our results for the exactness of seizure predictions against the best performing solutions from the Kaggle/Melbourne University seizure prediction challenge. This competition only focused on the solutions which had been evaluated by means of a single parameter, an Area Under the Curve (AUC). The winning team used 11 independent Machine Learning classifiers with more than 3000 manual EEG features; and obtained the mean AUC of 0.854 on the public test data and 0.791 on the private test data as presented in table 1 [22]. However, this approach is computationally expensive and not feasible in near real-time applications because of the need to extract many features from the EEG manualistically,Algorithmically, the top 5 Kaggle results utilized numerous hand-engineered features based on time, frequency, or time-frequency domain. To evaluate these features, Shown classifiers such as SVM, adaptive boost, trees, forests, and other classified were used.The best proposed solutions obtained an average AUC varying from 0.783 to 0.854 on the public test set and from 0.746 to 0.807 on the private test set[22].Further, [25] examined the possibility of improving the ensemble mixture using preictal probabilities from eight of the competition solutions. Regretfully, this method only gained a little improvement in prediction performance, with the average AUC as 0.815 on the public test set.

In two studies published most recently, [16] and [23], the use of convolutional neural networks (CNN) with different forms of input data provided a marked increase in seizure prediction performance. In [16], instead, CNNs were trained on the time frequency features, and in [23], raw EEG data was used as input for training. From the presented results, one can conclude that CNNs can be applied to a different set of options to predict seizures. However, the adopted approach in [23] employed training and testing the model for individual subjects, which poses questions on the generalisability of CNNs that directly work on raw EEG signal and hence needs further examination in subsequent research work. The AUC score is again good at 0.920 achieved by Varnosfaderani et al.[30] who trained a two-layer Long Short-Term Memory (LSTM) network. They extracted temporal features hand engineered that involved the mean, variance and peak to peak values of the epochs and spectral features consisting of the mean power across the first eight bands of the EEG signal. For demonstration purposes, our proposed MViT algorithm outperforms this case and provides an AUC score of 0.924 without the need for designing domain-specialized feature extraction.

Table 1 Comparison of previous seizure prediction methods with our MViT approach: Seizure prediction data set is from Melbourne University AES/MathWorks/NIH Seizure Prediction dataset.

Authors/Team	Year	EEG Features	Classifier	Sensitivity (%)	AUC Score (Public/Private)
Cook et al. [3]	2013	Signal energy	Decision tree, kNN	33.67	-
Karoly et al. [26]	2017	Signal energy, circadian profile	Logistic regression	52.67	-
Kiral-Kornek et al. [4]	2018	EEG Spectrogram, circadian profile	CNN	77.63	-
Kuhlmann,L. [19]	2018	Hurst exponent, spectral power, distribution attributes, fractal dimensions, AR error, and cross-frequency coherence	Extreme gradient boosting, kNN, SVM	-	0.853 / 0.807
Arete Associates [19]	2018	Correlation, entropy, zero-crossings, distribution statistics, and spectral power	Extremely randomized trees	-	0.783 / 0.799
GarethJones [19]	2018	Distribution statistics, spectral power, signal RMS, correlation, and spectral edge	SVM, tree ensemble	-	0.815 / 0.797
QingnanTang [19]	2018	Spectral power, spectral entropy, correlation, and spectral edge power	Gradient boosting, SVM	-	0.854 / 0.791
Nullset [19]	2018	Hjorth parameters, spectral power, spectral edge, spectral entropy, Shannon entropy, and fractal dimensions	Random Forest, adaptive boosting, gradient boosting	-	0.844 / 0.746
Reuben et al. [27]	2019	Preictal probabilities from the top 8 teams in [**?**]	MLP	-	0.815 / -
Varnosfaderani et al. [28]	2021	Temporal features, statistical moments, and spectral power	LSTM	86.80	0.920 / -
Hussein et al. [24]	2021	EEG Scalogram	SDCN	89.52	0.883 / -
Zhao et al. [29]	2022	Raw EEG	CNN	85.19–86.27	0.914–0.933 / -
Proposed Method	2022	EEG Scalogram	MViT	91.15	0.924 / -

5 Clinical Significance and Limitations

Clinical significance:This research also shows the importance of vision transformers in the simultaneous learning of features from multi-channel EEG data. The results also show the effectiveness of the outstanding EEG representations learnt by vision transformers in improving the seizure predictability from the two types of EEG records namely the surface and invasive EEG records.On the three independent public datasets, using the proposed MViT algorithm we obtained the maximum prediction sensitivity ranging from 90.28% to 99.80%, which opens the possibility for clinical trials of using EEG- based remote seizure warning systems. This particular MViT approach can predict the future seizures reliably and in the shortest time possible and most of the patients can take protective measures or give themselves quick acting anti-seizure medications when they are most vulnerable. In addition, this system could create the fundamentals of closed loop seizure interventional methods, designed to terminate seizure disorders in patients with intractable epilepsy.

Limitation:In the view of seizure-prediction outcomes previously reported for vision transformer concerning the applicability of based models, there are several issues arising from their use primarily within clinical framework. From [31], large-scale vision transformers have been shown to involve massive power and computational consumption. computational resources, the use of which is restricted by the amount of resources available in the devices they control.The mobile vision transformers in development today are also fields like brain computer interaction and prognosis of seizure. However, the process of how vision transformers arrive at a decision is not always well understood which is why knowing how such features as regions of an image that positively correlate with EEG classification impact decision-making can be difficult. Despite this, vision transformers have emerged as more effective horizon models than conventional convolutional and recurrent neural networks especially on issues to do with texture changes and contamination of data. They are also the better kinds since they are adaptable in actual practice and scenarios. Present studies are being directed

towards overcoming the computational hurdles in vision transformers primarily out of self-attention techniques[21].

6 Conclusions

In this study, we proposed a new algorithm named Multi-Channel Vision Transformer (MViT) that was used to predict seizures in epilepsy with high precision. The approach used was segmentation of the EEG signals with one data point per 10 seconds necessarily not overlapping, then converting these segments through Continuous Wavelet Transform(CWTS) into scalograms.The scalograms were segmented into fixed non-overlapping patches to ensure the model can meaningfully generalize the EEG features crucial to subtle seizure prediction using the MViT algorithm. MViT architecture may have several branches in which each branch takes a distinct EEG channel as an input. This design allows for therefore temporal-spectral features to be learned at the same time across multiple EEG channels. Finally, we proved the proposed MViT model outperforms many current and prior state-of-the-art architectures such as generic convolutional and recurrent neural networks-based seizure prediction methodologies comprehensively with the help of experiemental validation.

References

[1] Rogers, G.: R. appleton and a. marson 2009: Epilepsy: the facts, third edition. oxford, uk: Oxford university press. 186 pp, £9.99 paperback. isbn: 9780199233687. Primary Health Care Research 38; Development **11**(4), 413–413 (2010) https://doi.org/10.1017/S1463423610000241

[2] Acharya, U.R., Sree, S.V., Swapna, G., Martis, R.J., Suri, J.S.: Automated eeg analysis of epilepsy: a review. Knowledge-Based Systems **45**, 147–165 (2013)

[3] Cook, M.J., O'Brien, T.J., Berkovic, S.F., Murphy, M., Morokoff, A., Fabinyi, G., D'Souza, W., Yerra, R., Archer, J., Litewka, L., Hosking, S., Lightfoot, P., Ruedebusch, V., Sheffield, W.D., Snyder, D., Leyde, K., Himes, D.: Prediction of seizure likelihood with a long-term, implanted seizure advisory system in patients with drug-resistant epilepsy: a first-in-man study. The Lancet Neurology **12**(6), 563–571 (2013) https://doi.org/10.1016/S1474-4422(13)70075-9

[4] Kiral-Kornek, I., Roy, S., Nurse, E., Mashford, B., Karoly, P., Carroll, T., Payne, D., Saha, S., Baldassano, S., O'Brien, T., Grayden, D., Cook, M., Freestone, D., Harrer, S.: Epileptic seizure prediction using big data and deep learning: Toward a mobile system. EBioMedicine **27**, 103–111 (2018) https://doi.org/10.1016/j.ebiom.2017.11.032

[5] Shiao, H.-T., Cherkassky, V., Lee, J., Veber, B., Patterson, E.E., Brinkmann, B.H., Worrell, G.A.: Svm-based system for prediction of epileptic seizures from ieeg signal. IEEE Transactions on Biomedical Engineering **64**(5), 1011–1022 (2017) https://doi.org/10.1109/TBME.2016.2586475

[6] Ahmad, M.A., Khan, N.A., Majeed, W.: Computer assisted analysis system of electroencephalogram for diagnosing epilepsy. In: 2014 22nd International Conference on Pattern Recognition, pp. 3386–3391 (2014). https://doi.org/10.1109/ICPR.2014.583

[7] Zabihi, M., Kiranyaz, S., Ince, T., Gabbouj, M.: Patient-specific epileptic seizure detection in long-term EEG recording in paediatric patients with intractable seizures, pp. 7–06. https://doi.org/10.1049/cp.2013.2060 . https://digital-library.theiet.org/doi/abs/10.1049/cp.2013.2060

[8] Williamson, J.R., Bliss, D.W., Browne, D.W., Narayanan, J.T.: Seizure prediction using eeg spatiotemporal correlation structure. Epilepsy Behavior **25**(2), 230–238 (2012) https://doi.org/10.1016/j.yebeh.2012.07.007

[9] Khan, H., Marcuse, L., Fields, M., Swann, K., Yener, B.: Focal onset seizure prediction using convolutional networks. IEEE Transactions on Biomedical Engineering **65**(9), 2109–2118 (2018) https://doi.org/10.1109/TBME.2017.2785401

[10] Truong, N.D., Nguyen, A.D., Kuhlmann, L., Bonyadi, M.R., Yang, J., Ippolito, S., Kavehei, O.: Convolutional neural networks for seizure prediction using intracranial and scalp electroencephalogram. Neural Networks **105**, 104–111 (2018) https://doi.org/10.1016/j.neunet.2018.04.018

[11] Wang, G., Wang, D., Du, C., Li, K., Zhang, J., Liu, Z., Tao, Y., Wang, M., Cao, Z., Yan, X.: Seizure prediction using directed transfer function and convolution neural network on intracranial eeg. IEEE Transactions on Neural Systems and Rehabilitation Engineering **28**(12), 2711–2720 (2020) https://doi.org/10.1109/TNSRE.2020.3035836

[12] Goldberger, A.L., Amaral, L.A.N., Glass, L., Hausdorff, J.M., Ivanov, P.C., Mark, R.G., Mietus, J.E., Moody, G.B., Peng, C.-K., Stanley, H.E.: Physiobank, physiotoolkit, and physionet. Circulation **101**(23), 215–220 (2000) https://doi.org/10.1161/01.CIR.101.23.e215 https://www.ahajournals.org/doi/pdf/10.1161/01.CIR.101.23.e215

[13] Ozcan, A.R., Erturk, S.: Seizure prediction in scalp eeg using 3d convolutional neural networks with an image-based approach. IEEE Transactions on Neural Systems and Rehabilitation Engineering **27**(11), 2284–2293 (2019) https://doi.org/10.1109/TNSRE.2019.2943707

[14] Liu, C.-L., Xiao, B., Hsaio, W.-H., Tseng, V.S.: Epileptic seizure prediction with multi-view convolutional neural networks. IEEE Access **7**, 170352–170361 (2019) https://doi.org/10.1109/ACCESS.2019.2955285

[15] Qi, Y., Ding, L., Wang, Y., Pan, G.: Learning robust features from nonstationary brain signals by multiscale domain adaptation networks for seizure prediction. IEEE Transactions on Cognitive and Developmental Systems **14**(3), 1208–1216

(2022) https://doi.org/10.1109/TCDS.2021.3100270

[16] Hussein, R., Lee, S., Ward, R., McKeown, M.J.: Semi-dilated convolutional neural networks for epileptic seizure prediction. Neural Networks **139**, 212–222 (2021) https://doi.org/10.1016/j.neunet.2021.03.008

[17] Lian, Q., Qi, Y., Pan, G., Wang, Y.: Learning graph in graph convolutional neural networks for robust seizure prediction. Journal of Neural Engineering **17**(3), 035004 (2020) https://doi.org/10.1088/1741-2552/ab909d

[18] Shoeb, A.H.: Application of machine learning to epileptic seizure onset detection and treatment. PhD thesis, Massachusetts Institute of Technology (2009)

[19] Kuhlmann, L., Karoly, P., Freestone, D.R., Brinkmann, B.H., Temko, A., Barachant, A., Li, F., Titericz, J. Gilberto, Lang, B.W., Lavery, D., Roman, K., Broadhead, D., Dobson, S., Jones, G., Tang, Q., Ivanenko, I., Panichev, O., Proix, T., Náhlík, M., Grunberg, D.B., Reuben, C., Worrell, G., Litt, B., Liley, D.T.J., Grayden, D.B., Cook, M.J.: Epilepsyecosystem.org: crowdsourcing reproducible seizure prediction with long-term human intracranial eeg. Brain **141**(9), 2619–2630 (2018) https://doi.org/10.1093/brain/awy210 https://academic.oup.com/brain/article-pdf/141/9/2619/25590596/awy210.pdf

[20] Dosovitskiy, A.: An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929 (2020)

[21] Vaswani, A.: Attention is all you need. Advances in Neural Information Processing Systems (2017)

[22] Brinkmann, B.H., Wagenaar, J., Abbot, D., Adkins, P., Bosshard, S.C., Chen, M., Tieng, Q.M., He, J., Muñoz-Almaraz, F.J., Botella-Rocamora, P., Pardo, J., Zamora-Martinez, F., Hills, M., Wu, W., Korshunova, I., Cukierski, W., Vite, C., Patterson, E.E., Litt, B., Worrell, G.A.: Crowdsourcing reproducible seizure forecasting in human and canine epilepsy. Brain **139**(6), 1713–1722 (2016) https://doi.org/10.1093/brain/aww045

[23] Zhao, S., Yang, J., Sawan, M.: Energy-efficient neural network for epileptic seizure prediction. IEEE Transactions on Biomedical Engineering **69**(1), 401–411 (2022) https://doi.org/10.1109/TBME.2021.3095848

[24] Hussein, R., Lee, S., Ward, R., McKeown, M.J.: Epileptic seizure prediction: A semi-dilated convolutional neural network architecture. In: 2020 25th International Conference on Pattern Recognition (ICPR), pp. 5436–5443 (2021). https://doi.org/10.1109/ICPR48806.2021.9412384

[25] Reuben, C., Karoly, P., Freestone, D.R., Temko, A., Barachant, A., Li, F., Titericz Jr., G., Lang, B.W., Lavery, D., Roman, K., Broadhead, D., Jones, G., Tang, Q., Ivanenko, I., Panichev, O., Proix, T., Náhlík,

M., Grunberg, D.B., Grayden, D.B., Cook, M.J., Kuhlmann, L.: Ensembling crowdsourced seizure prediction algorithms using long-term human intracranial eeg. Epilepsia **61**(2), 7–12 (2020) https://doi.org/10.1111/epi.16418 https://onlinelibrary.wiley.com/doi/pdf/10.1111/epi.16418

[26] Karoly, P.J., Ung, H., Grayden, D.B., Kuhlmann, L., Leyde, K., Cook, M.J., Freestone, D.R.: The circadian profile of epilepsy improves seizure forecasting. Brain **140**(8), 2169–2182 (2017) https://doi.org/10.1093/brain/awx173 https://academic.oup.com/brain/article-pdf/140/8/2169/24175506/awx173.pdf

[27] Korshunova, I., Kindermans, P.-J., Degrave, J., Verhoeven, T., Brinkmann, B.H., Dambre, J.: Towards improved design and evaluation of epileptic seizure predictors. IEEE Transactions on Biomedical Engineering **65**(3), 502–510 (2018) https://doi.org/10.1109/TBME.2017.2700086

[28] Chen, R., Parhi, K.K.: Seizure prediction using convolutional neural networks and sequence transformer networks. In: 2021 43rd Annual International Conference of the IEEE Engineering in Medicine Biology Society (EMBC), pp. 6483–6486 (2021). https://doi.org/10.1109/EMBC46164.2021.9629732

[29] Ma, X., Qiu, S., Zhang, Y., Lian, X., He, H.: Predicting epileptic seizures from intracranial eeg using lstm-based multi-task learning. In: Chinese Conference on Pattern Recognition and Computer Vision (PRCV), pp. 157–167 (2018). Springer

[30] Varnosfaderani, S.M., Rahman, R., Sarhan, N.J., Kuhlmann, L., Asano, E., Luat, A., Alhawari, M.: A two-layer lstm deep learning model for epileptic seizure prediction. In: 2021 IEEE 3rd International Conference on Artificial Intelligence Circuits and Systems (AICAS), pp. 1–4 (2021). https://doi.org/10.1109/AICAS51828.2021.9458539

[31] Khan, S., Naseer, M., Hayat, M., Zamir, S.W., Khan, F.S., Shah, M.: Transformers in vision: A survey. ACM Comput. Surv. **54**(10s) (2022) https://doi.org/10.1145/3505244